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THORACOSTOMY SKILL GUIDE

SFM CSOG 003 'MANAGEMENT OF CHEST INJURIES' 2022. MANAGE VENTILATORY FAILURE AND IDEALLY SHOULD BE PERFORMED WITH CONCOMITANT VOLUME RESUSCITATION IN SHOCKED PATIENTS.

IS PATIENT SPONTANEOUSLY BREATHING?

NO

YES

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PLACE ADVANCED AIRWAY & PERFORM POSITIVE PRESSURE VENTILATION (PPV). PERFORM OPEN/FINGER THORACOSTOMY.

PERFORM TUBE THORACOSTOMY WITH EQUIPMENT READY FOR TRAUMATIC CARDIAC ARREST

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OPEN/FINGER THORACOSTOMY - PATIENT MUST HAVE PPV

PREPARATION

KIT DUMP: CHLORAPREP WIPES, SCALPEL, SPENCER WELLS & CHEST SEAL

POSITION PATIENT: 360 ACCESS WITH ARM(S) ABDUCTED [HUMERAL IO??]

IDENTIFY & CLEAN LANDMARKS: TRIANGLE OF SAFETY AIM FOR 4TH IC MAAL

PROCEDURE

MAKE A ~4cm INCISION ON THE 4TH IC MAAL OVER THE 5TH RIB AND FOLLOWING THE CURVE OF RIB. DO NOT INCISE PAST THE MUSCLE. IF ABNORMAL CHEST ANATOMY, ENSURE INCISION IS ABOVE NIPPLE LINE.

BLUNT DISSECT THE MUSCULATURE INTO THE PLEURA WITH SPENCER WELLS FORCEPS WITH A FINGER 3-4cm FROM THE TIP TO ACT AS A STOP. A RELEASE OF AIR SHOULD BE FELT AND EXPECT BLEEDING. OPEN THE FORCEPS FULLY THEN CLOSE, ROTATE 90° THEN OPEN TO WIDEN TRACT.

REMOVE FORCEPS AND INSERT GLOVED FINGER TO PERFORM A FINGER SWEEP, CHECKING IF LUNG IS EXPANDED (UP) OR COLLAPSED (DOWN). LUNG SHOULD BE BEGIN TO EXPAND ON RELEASE OF PNEUMOTHORAX.

CONSIDER THE OTHER SIDE, ESPECIALLY IN TRAUMATIC CARDIAC ARREST AND/OR DIFFICULTY VENTILATING. DO NOT FORGET. IF PATIENT IMPROVES THEN PHEA SHOULD BE CONSIDERED.

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TUBE THORACOSTOMY (CHEST DRAIN) - PPV FOR BILATERAL THORACOSTOMIES

PREPARATION

KIT DUMP: CHLORAPREP WIPES, CHEST DRAIN KIT & CHEST SEALS

POSITION PATIENT: SEMI-RECUMBENT, 360 ACCESS WITH ARM(S) ABDUCTED

IDENTIFY & CLEAN LANDMARKS: TRIANGLE OF SAFETY AIM FOR 4TH IC MAAL

PROCEDURE

ADMINISTER LOCAL ANAESTHETIC (20ml LIDOCAINE 1%) MAX OF 15ml IN A FANNED PATTERN AROUND AREA OF DRAIN INSERTION THEN INSERT NEEDLE INTO PLUERAL SPACE, ASPIRATE THE AIR AND SLOWLY WITHDRAW UNTIL ASPIRATION STOPS. INJECT 5ml OF LOCAL INTO PARIETAL PLEURA ALLOWING 3-5 MINUTES FOR EFFECT. CONSIDER PROCEDURAL SEDATION.

MAKE A ~4cm INCISION ON THE 4TH ICMAAL OVER THE 5TH RIB AND FOLLOWING THE CURVE OF RIB. DO NOT INCISE PAST THE MUSCLE. IF ABNORMAL CHEST ANATOMY, ENSURE INCISION IS ABOVE NIPPLE LINE.

BLUNT DISSECT THE MUSCULATURE INTO THE PLEURA WITH SPENCER WELLS FORCEPS WITH A FINGER 3-4cm FROM THE TIP TO ACT AS A STOP. A RELEASE OF AIR SHOULD BE FELT AND EXPECT BLEEDING. OPEN THE FORCEPS FULLY THEN CLOSE, ROTATE 90° THEN OPEN TO WIDEN TRACT.

REMOVE FORCEPS AND INSERT GLOVED FINGER TO PERFORM A FINGER SWEEP, CHECKING IF LUNG IS EXPANDED (UP) OR COLLAPSED (DOWN).

SIMULTANEOUSLY REMOVE FINGER AND FULY INSERT CHEST TUBE WITH INTRODUCER, AIMING SLIGHTLY POSTERIORLY. SECURE TUBE WITH MODIFIED JO'BURG TECHNIQUE (SUTURES) AND APPLY OCCLUSIVE DRESSING.

APPLY CHEST SEALS IF UNABLE TO PERFORM JO'BURG SUTURING.

IF BILATERAL THORACOSTOMIES IS REQ'D, ENSURE PPV IS AVAILABLE AND PERFORM ONE AT A TIME. MONITOR FOR ARREST. NEEDLE DECOMPRESSION CAN BE DONE BY OTHER TRAINED MEDIC